

Pandas Cheatsheet — 40 Essential Functions (PDF-Ready)

Data Loading & Inspection

```
pd.read_csv("file.csv") – Load CSV
pd.read_json("data.json") – Load JSON
df.info() – Schema + nulls
df.head(10) – Preview rows
df.sample(5) – Random sample
df.shape – Rows/columns
df.describe() – Summary stats
```

Selecting & Filtering

```
df['col'] – Select column
df[['a','b']] – Select multiple columns
df[df['age'] > 30] – Boolean filter
df[(df['age']>18) & (df['country']=='US')] – Multi-condition
df.query("country=='US'") – SQL-style filter
df[df['age'].between(20,30)] – Range filter
```

Cleaning & Transforming

```
df.dropna(subset=['id']) – Drop missing
df.fillna(0) – Fill missing
df.drop_duplicates(subset=['id']) – Deduplicate
df.duplicated() – Check duplicates
df.columns.difference(['id']) – Exclude columns
df.reset_index(drop=True) – Reset index
df.rename(columns={'a':'A'}) – Rename
df.drop(columns=['col']) – Drop column
df.replace({'A':1}) – Replace values
np.where(df['x']>0,1,0) – Conditional column
```

Type Handling

```
df['id'].astype('Int64') – Cast type
pd.to_datetime(df['ts']) – To datetime
df['ts'].dt.date – Extract date
df['ts'].dt.year – Extract year
```

Aggregation & Grouping

```
df.groupby('cat')['sales'].sum() – Group + sum
df.groupby('id').agg({'a':'sum','b':'count'}) – Multi-agg
df['id'].count() – Count
df['user_id'].nunique() – Unique count
df['country'].value_counts() – Frequency
df.sort_values('sales', ascending=False) – Sort
```

Joining & Combining

```
pd.merge(a, b, on='id', how='left') – SQL join
df1.join(df2, how='left') – Index join
pd.concat([df1, df2]) – Stack DataFrames
```

Advanced Operations

```
df['ma7'] = df['sales'].rolling(7).mean() – Rolling window
```

```
df['r'] = df.groupby('cat')['sales'].rank() – Rank  
df.pivot_table(values='sales', index='cat', columns='region') – Pivot  
df['x2'] = df['x'].apply(lambda x: x*2) – Apply  
df['email'].str.lower() – String ops  
df.loc[df['age']<0, 'age'] = 0 – Update rows
```

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